

Bachelor in Computer Applications

Course Title: **Data Structure and Algorithms**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. How stack is used in recursion? Explain different stack operations. Explain algorithm to convert an infix expression to postfix using stack.
2. Explain complete binary tree with example. Starting with an empty binary search tree, show the effect of successively adding the following elements: 47, 50, 25, 27, 17, 61, 5, and 26. Also, traverse the resulting tree in pre-order, in-order, and post-order.
3. Explain quick sort algorithm. Use this algorithm to sort the numbers 35, 82, 18, 54, 13, 31, 20, 69, and 19.
4. Define ADT. Explain array as an ADT.
5. What is time complexity? Explain big oh notation with example.
6. Explain priority queue with example. What is circular queue?
7. What are the benefits of using recursion? Write a recursive function to find nth Fibonacci number.
8. Explain singly linked list with example. Compare singly linked list with doubly linked list.
9. Explain different applications of binary tree.
10. Explain collision and collision resolution in hashing. What is double hashing?
11. Explain adjacency matrix representation of graphs with example.
12. Write short notes on:
a. Linear Search
b. Minimum Spanning Tree